

MSBA Job Matching - Final Technical Summary

Problem and Production Objective

The system supports a graduate business career advisor screening opportunities for MSBA international students. The production objective is faster first-pass prioritization with inspectable evidence, while keeping work-authorization and final forwarding decisions under human review.

System Architecture

Stage	Production behavior
Input	Accept an advisor query or raw posting text through the CLI
Retrieve	Search 80,000 real training postings and return top-6 evidence
Decide	Apply transparent four-label role-fit scoring
Generate	Create a recommendation only from retrieved fields
Govern	Surface explicit authorization text; escalate missing evidence

Measured Results

System	Rows	Accuracy	Macro F1	Interpretation
M2 TF-IDF	20k	0.793	0.785	Static baseline
M3 PEFT LoRA	4k	0.875	0.873	Best classifier
Final system	4k	0.800	0.801	Evidence + guardrails

On the same balanced 4,000-row subset, final macro F1 is 0.801 versus 0.791 for the comparable baseline. Retrieval success is 100.0%, and support hit@5 is 90.0%. The M3 LoRA model remains the best pure classifier; the final system is selected for the product because it returns inspectable evidence and a governance response.

Real Runs and Failure Handling

Case	Predicted	Top evidence	Decision support
entry level analytics	high_fit	Data Analyst (high_fit)	Forward candidates after advisor review
senior engineering filter	low_fit	Adidas Recruitment 2023 - 2+Years Experience Require (medium_fit)	Hold a role that is too senior or engineering-heavy
authorization missing	unclear	Jr. Data Analyst (No OPT/CPT) (high_fit)	Escalate missing CPT/OPT or sponsorship evidence
thin metadata	unclear	ESM Data Analytics Internship - Summer 2023 (high_fit)	Investigate an incomplete posting

Governance

The casebook tests the highest-impact failure mode directly. All 4 cases returned evidence, explicit authorization language was surfaced when present, and the check found zero unsupported authorization claims. This does not prove general safety, but it verifies the intended guardrail on the submitted demonstrations.

Weak labels remain the largest modeling limitation. Medium-fit and unclear postings are often pushed to low fit, so the production recommendation is advisor-confirmed prioritization rather than automated rejection. Future private data must also be de-identified, access-controlled, and retained only as long as necessary.

Business Recommendation

Pilot the system with de-identified Career Center postings and measure high-fit precision, advisor review time, override rate, and authorization-extraction accuracy. Ship the current version only as a decision-support prototype. Add a separate authorization extractor after reliable source text and human-reviewed labels are available.